THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET C

Chapters 3 & 4: **The World of Numbers, Exploring Algebraic Identities** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The decimal expansion of $\frac{4}{15}$ is: 1

(a) non-terminating non-repeating **(b)** non-terminating repeating
(c) terminating **(d)** cannot be determined

2. The number $0.\overline{6}$ in the form $\frac{p}{q}$ is: 1

(a) 1 **(b)** $\frac{2}{3}$ **(c)** $\frac{3}{2}$ **(d)** $\frac{23}{30}$

3. The number $0.1\overline{6}$ in the form $\frac{p}{q}$ is: 1

(a) $\frac{4}{15}$ **(b)** 6 **(c)** $\frac{1}{3}$ **(d)** $\frac{1}{6}$

4. The number $0.333\ldots$ is: 1

(a) not a real number **(b)** irrational
(c) neither rational nor irrational **(d)** rational

5. The number $3+\sqrt{5}$ is: 1

(a) rational **(b)** not a real number
(c) neither rational nor irrational **(d)** irrational

6. Expand $(4x+3)^2$: 1

(a) $16x^2+24x-9$ **(b)** $16x^2+9$ **(c)** $16x^2+12x+9$ **(d)** $16x^2+24x+9$

7. Expand $(3x-4)^2$: 1

(a) $9x^2-16$ **(b)** $9x^2-24x+16$ **(c)** $9x^2-12x+16$ **(d)** $9x^2+24x+16$

8. The value of 99^2 using a suitable identity is: 1

(a) 9821 **(b)** 9791 **(c)** 9801 **(d)** 9901

9. The expression $36x^2+12x+1$ can be factorised as: 1

(a) $(6x-1)^2$ **(b)** $(6x+1)^2$ **(c)** $(6x+2)^2$ **(d)** $(36x+1)^2$

10. $(x+7)(x+8)$ equals: 1

(a) $x^2+56x+15$ **(b)** x^2-x+56 **(c)** $x^2+15x+56$ **(d)** $x^2+15x-56$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Locate $\frac{3}{4}$ on the number line by describing the steps. 2

12. Show that $0.\overline{9}=1$ using algebra. 2

13. Evaluate 119^2 using $(a+b+c)^2$. 2

14. Evaluate 23×17 using a suitable identity. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A tailor has 15 m of silk and a kurta needs 2 m. How many kurtas can be made and how much silk is left? Express the number of kurtas as a rational number. 3

16. The breadth of a pool is 4 m less than its length and its area is 96 m². Find its dimensions. 3

17. Prove that $5 - \sqrt{3}$ is irrational, given that $\sqrt{3}$ is irrational. 3

OR

Show that $\frac{2}{\sqrt{3}}$ is irrational given that $\sqrt{3}$ is irrational. 3

18. Factorise $r^2 - r - 42$ and $x^2 + 7x + 12$. 3

OR

Fill the blanks: $(px+a)(qx+b) = (__)x^2 + (__)x + __$. Verify with $(2x+3)(3x+1)$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** Brahmagupta's rules: fortune + fortune = fortune; debt + debt = debt; debt × fortune = debt; debt × debt = fortune. Using these, 5

(i) compute $(-3) \times 4$ and $(-3) \times (-4)$;

(ii) a spice trader borrows ₹850, earns ₹1,200 and loses ₹450 — find his final standing;

(iii) explain $(-3) \times (-4) = 12$ in terms of debt.

OR

(i) Find the rational number x such that $x\left(\frac{1}{3} + \frac{1}{6}\right) = \frac{9}{2}$. 5

(ii) Verify $\frac{1}{2}\left(\frac{3}{4} + \frac{1}{3}\right) = \frac{1}{2} \times \frac{3}{4} + \frac{1}{2} \times \frac{1}{3}$.

20. Find the values of $(i)\ 1104^2$ and $(ii)\ 198^2$ using suitable identities. 5

(iii) Which identity makes each easier?

OR

Case study: A rectangular garden has length $(2x+3)$ m and breadth $(3x+1)$ m. 5

(i) Write and simplify its area.

(ii) Find the area when $x = 4$.

(iii) A square flower bed has side $(x+2)$; expand its area and compare with a $(x+3) \times (x+1)$ plot.

END OF QUESTION PAPER — **Total: 40 Marks**

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